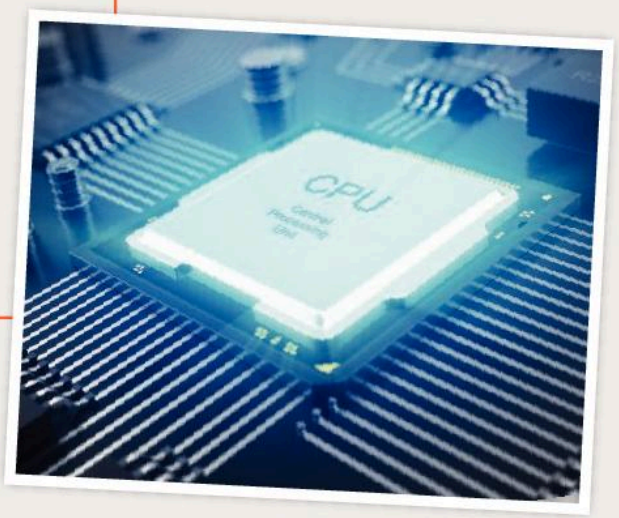
1 The nature of technology: Computers around us

You will learn

- that computer power improves how devices such as TVs and cars work
- how computers and technology improve the way people work
- about computer storage and why it is important.



Computer power is used to improve the way that devices work at home, at work and at school. TVs, cars, mobile devices, smartphones and fridges are all powered by computers. Computers are changing the way we live.



Talk about...

If all computers disappeared tomorrow...

- what would you miss about computers in your home life?
- what would you miss about computers at school?



Learning outcomes: Describe how computers may be used in the world of work; Identify a range of modern devices which contain computer processors (for example, embedded processors); Describe what storage is and why it is important

Class activity

Make a list of the types of computers you use at home and at school. For example, do you use a tablet computer? Do you use different types of computers to do different things?

microprocessor
robot sensors
storage drive
data file back-up file
flash drive



Digital citizen of the future

Computers can improve the way we learn, work and enjoy our spare time. However, not everyone can afford to buy a computer. Some charities collect computers that are no longer used. These computers are given to people who cannot afford to buy their own. Would you donate a computer you no longer use?

Did you know?

In 2015, scientists at Michigan University created Michigan Micro Mote, the world's smallest computer. The Mote measures 2mm × 2mm × 4mm. That's smaller than a grain of rice! The Mote is being used in medical implants and in driverless cars.

1.1 The power of computers

In this lesson

You will learn:

- how computers help us to work
- how important a microprocessor is to a computer
- that microprocessors are small enough to be used in other devices.

Spiral back

Last year, you learned that there are different types of computer. Desktop computers, laptops, tablets and smartphones are all types of computer. Now you will learn about how we use computers to help us learn, work and enjoy our free time.

How computers help us to work

You use computers at home and at school to help you to do your school work. You can use a computer to research information for a project. You use a word processor to create homework that looks good and is easy to read. Computers make it easy for you to make changes to documents.

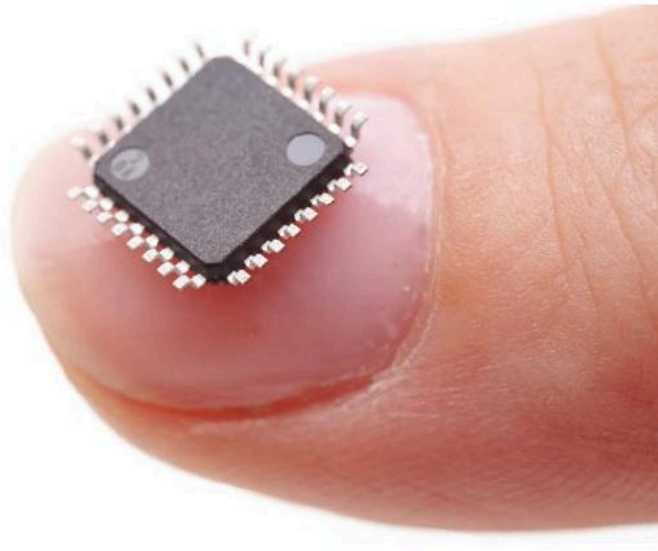
You use computers to communicate and share ideas with other people. You use email to share your documents with classmates and your teacher.

Computers help people to do their jobs.

- People use word processors to **produce documents**, such as reports, plans and letters to customers. Spreadsheets show how much money a business is making. Attractive presentations give information to staff and customers.
- Computers help people to **find information** they need to do their jobs. Computers can store a lot of data. You can find information quickly.
- Computers send emails and messages to customers and work colleagues. Voice and video conversations mean people can **communicate** wherever they are in the world.



Computers helps us to work quickly and accurately. A computer is a powerful tool.



What makes a computer so powerful?

A computer has many parts inside. The most important part inside a computer is called the microprocessor.

The **microprocessor** is the computer's brain. It does all the important work inside the computer. Whenever you use a computer, everything you see on the screen or hear through the speakers has been created by the microprocessor. A microprocessor is so small it will fit on the tip of your finger.

Where else can you find microprocessors?

Microprocessors are not just used in computers – they are everywhere! A microprocessor is small enough to fit into almost any device or machine. Microprocessors are used in the home in devices like televisions. Microprocessors are used in the workplace in **robots** and other machines. Microprocessors make devices more powerful and easier to use.



Activity

List the types of computer you use at home or at school. Which type of computer do you prefer to use for your school work? Why?

Think again

Talk to a family member who is in work. What do they use computers for at work? How do computers make it easier for them to do their job?



Extra challenge

List three ways that you use computers to help you do school work. For each one, say how you would complete the work without a computer. How does using a computer make it easier to do your school work?

1.2 Microprocessors at home

In this lesson

You will learn:

- how microprocessors improve how devices work in the home and in a car.

Microprocessors in the home

TV

Microprocessors let us pause programmes or record them to watch later. TVs have remote controls to change channels and adjust the volume. You can control some TVs with your voice. These functions are controlled by microprocessors.



Washing machine

Microprocessors make sure you can wash your clothes without damaging them. They control the amount of time that the clothes are washed. They also control the speed of the spin dryer.

Fridge

Sometimes, a device with a microprocessor is also connected to the internet. A device which is connected to the internet is a **smart device**. A smart refrigerator warns us if food is out of date. You can check what is in your smart refrigerator using your smartphone while you are at the supermarket.



Home systems

Microprocessors are used in home alarm and heating systems. Sensors can be installed in rooms to detect movement. A microprocessor uses information from the sensors to sound an alarm or turn on lights and heating when someone enters a room.

Microprocessors in the car

A modern car can have as many as 60 **embedded microprocessors**. Each microprocessor does a special job. For example, one microprocessor will manage how the car uses fuel. Another will operate the car radio.



Microprocessors make cars:

- easier to drive, for example, satellite navigation helps drivers get to where they want to go
- more comfortable for passengers, for example, the temperature in the car is kept at the right level
- safer, for example, brakes can be applied if the car gets too close to another vehicle.



Activity

Explore your home and make a list of the items you can find that use microprocessor technology. If a device has a display like the one in the photo of the washing machine, it probably has an embedded microprocessor.



Extra challenge

Search the internet to find out more about smart refrigerators. Make a list of the things a smart refrigerator can do.



Explore more

Pick a device from the list you made in the activity. Talk to a parent or grandparent about that device. Is it better than similar devices they used in the past? How has technology changed since your parent or grandparent was your age?

1.3 Microprocessors at work

In this lesson

You will learn:

- how microprocessors are used to improve the way people work.



Doctors

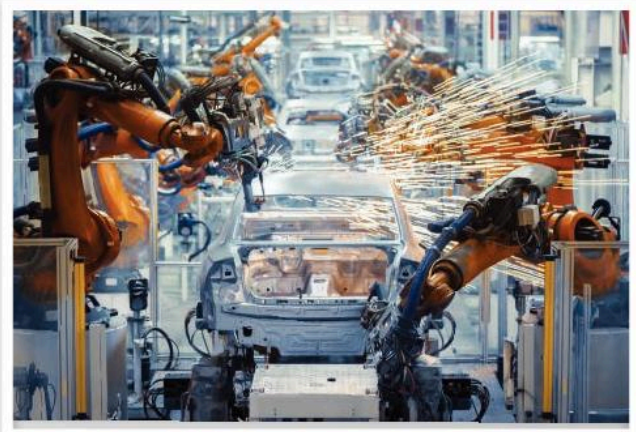
Doctors diagnose illness using machines that contain microprocessors. Two important machines are magnetic resonance imaging (MRI) and computerised axial tomography (CAT) scanners. MRI and CAT scanners create a 3D picture of the inside of a patient. A doctor can use the picture to see problems.

Technology also helps patients to recover after treatment. Patient monitoring systems check blood pressure, temperature and pulse rate. Nurses get an early warning if a patient needs attention.

Being a doctor is a complex job. There is a lot to know about many illnesses. Doctors use the internet for research. They also use the internet to contact experts. Sometimes those experts work in another country.

Manufacturing

A modern car factory is full of technology. Robots are used to assemble cars. A car is moved automatically from one part of the factory to another. At each stop, robots complete work on part of the car. Robots do work that used to be done by humans.



Robots are used to manufacture many other goods. They are used for simple repetitive jobs. Robots are also used to do jobs that are dangerous for humans. For example, the police use robots to investigate packages that might contain explosives.

Retail



Supermarkets and other large shops depend on technology. At the supermarket checkout, goods are scanned using a barcode reader. This creates a detailed receipt for the customer. Information gathered at the checkout is used by the supermarket to decide what items they need to order from their suppliers. Customers pay for their shopping using debit and credit cards. Money is transferred automatically from the customer to the shop.

Many people now use the internet to do their shopping. Orders made on the internet are processed quickly by computer systems. Goods are often delivered on the day after an order is made. Some people worry about the effect internet shopping has on shops in town centres.



Activity

Working in a group, interview your teacher to find out how using technology improves their job. Work together to write a list of three questions before you start the interview.



Extra challenge

Jobs that used to be done by humans are being replaced by automation and robots. Talk about this with classmates and your family. What are the good and bad things about automation?

Think again

Do you think your teacher could be replaced by a robot? Give reasons for your answer. How would you feel about travelling to school in a bus driven by a robot?

1.4 Computer storage

In this lesson

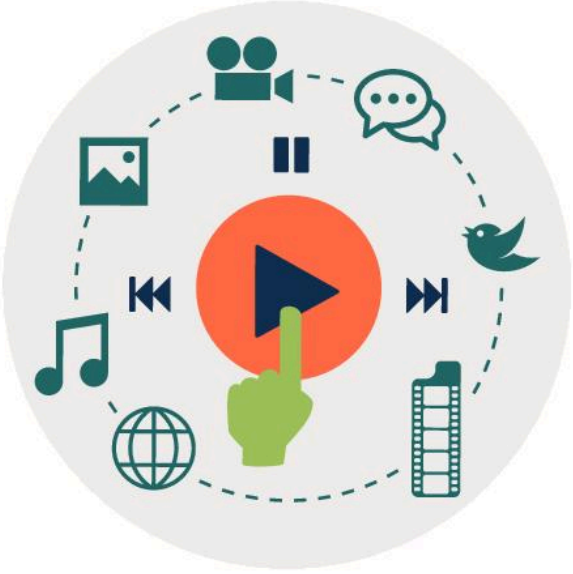
You will learn:

- about computer storage
- that computer storage is an important component of a computer system
- why computer storage is important.

Data files

Different types of media can be created and used on a computer. The media types used on computers include:

- text documents (for example, homework or assignments)
- pictures and photographs
- videos
- music



All types of media used on a computer are stored in files. A file stored on a computer is called a **data file**.

Computer storage

When you create work on a computer, it is important to save it correctly. Work saved in a data file can be used over and over again. If your work is not saved correctly it can be lost.

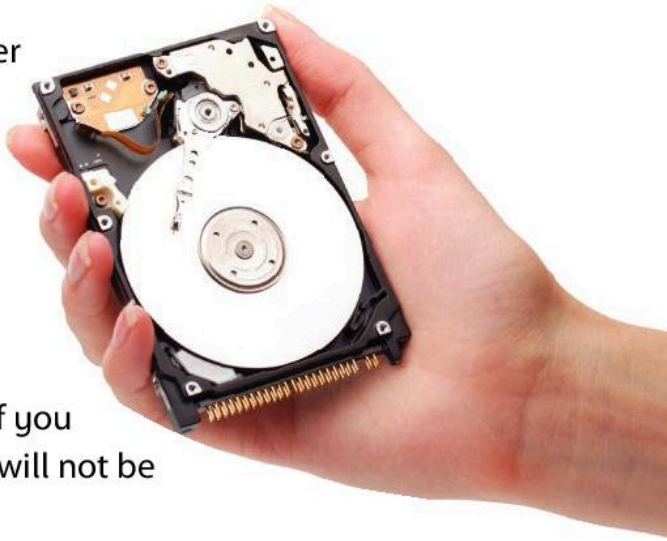
All computers have a place to store data files. It is called computer storage.

Where data files are stored

A computer uses a device called a **storage drive** to save files. Every computer has a storage drive built into the case. A storage drive can save a large amount of information.

For example, a storage drive used in a desktop computer can store:

- 150,000 photographs or
- 8500 hours of music or
- 300,000 books.



Network storage

You will probably not always use the same computer. If you save work onto the storage drive of one computer, you will not be able to use that file if you move to another computer.

Computers in schools and offices are usually connected. We call a group of connected computers a **network**. A network has its own storage drive. Everyone using a network has a special area on the **network storage** drive where they can save their work. If you save your work on your school network drive, you can use it on any computer that is connected to your school network.

Flash drives

A **flash drive** is a small portable storage device. It is small enough to fit in your pocket and can be used to move files from one computer to another. A flash drive is plugged into a computer so that a data file can be saved to it.





Activity

How could you use a flash drive to move homework from a computer at home to a computer at school? Write your answer as a list of steps.



Extra challenge

You have learned that a typical storage drive can hold 8500 hours of music. How many days would it take to play that music? (Hint: there are 24 hours in a day.)

Think again

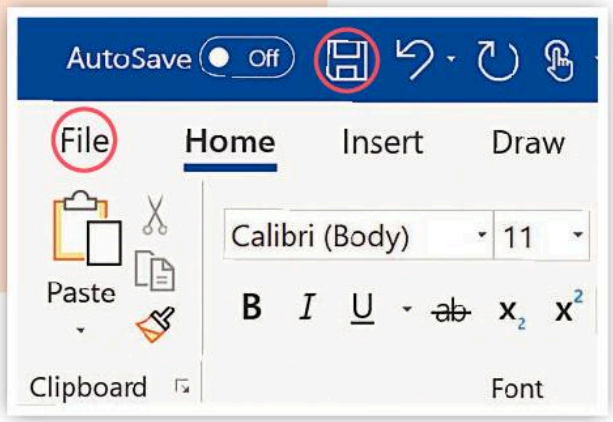
What problems will you have if you don't save your work properly?

1.5 How to use and store your files

In this lesson

You will learn:

- how to use files and folders
- how to store files.



Using and storing files

How to save a file

If you are working in an application such as a word processor, there are two ways to save a file. You can use the File menu or click the Save icon. If your file already has a name, it will be saved with the same name. If it is a new file, you need to enter a file name before you can save the file. You will learn more about choosing a good name for your file in Lesson 1.6.

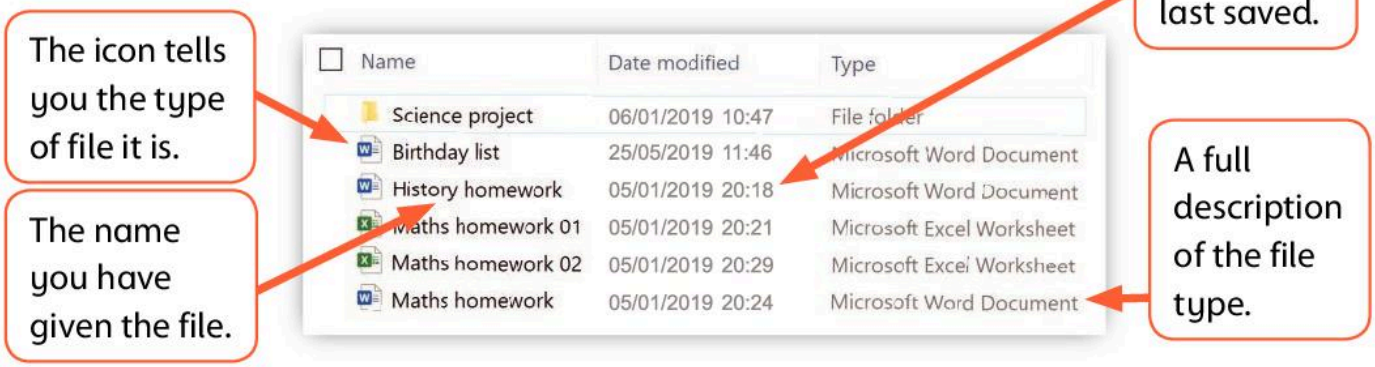
How to copy a file

You can reduce the risk of losing important work by making a copy of the file. The copy is called a **back-up file**.

- Find the file in File Explorer.
- Right click your mouse over the file, then select Copy from the menu that appears.
- Move to the folder where you want to save your back-up. Right click your mouse and select Paste from the menu.

How to find a file

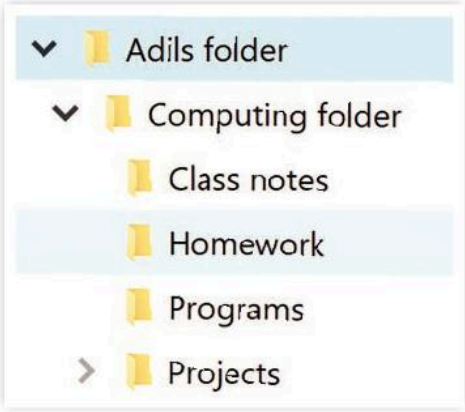
File Explorer gives information to help you find a file. The image shows a list of files and information that can help you identify them.



Using folders

Folders can help you to organise your work. You can organise your work by subject or type. For example, you can set up a folder for your computing work.

Inside the computing folder, you can set up folders to contain different types of work. For example, you could set up folders for homework, class notes, projects and programs.



How to create a folder

In File Explorer, create a new folder by following these steps.

1. Go to the area where you want to add a new folder.
2. Right click using your mouse.
3. Select New from the menu that appears, then click Folder.
4. Enter a name for the new folder.



Activity

Look at the list of files at the bottom of page 14. Sunny is searching for a file she knows she created using Word on 5 January 2019. Which files in the list can she dismiss? For each file, say why.



Extra challenge

Open your work area in File Explorer, either on a school computer or at home. Create a folder called 'Back-ups'. Copy a file to your 'Back-ups' folder.

Think again

How do you organise your data files at home and at school? What can you do to improve the way you use folders and file names?

1.6 Tips for saving files

In this lesson

You will learn:

- why it is important to save your work correctly
- some rules to help you remember to save your work correctly.

Why saving your work is important

Saving work correctly is important. Failing to save work correctly could mean:

- you will lose work and must start again
- you may not be able to find work when you need it again.

Rules to help you when you are saving files

Rule 1: Use a helpful file name

When you save a file, you must give it a name. Choose a name that describes the information in your file.

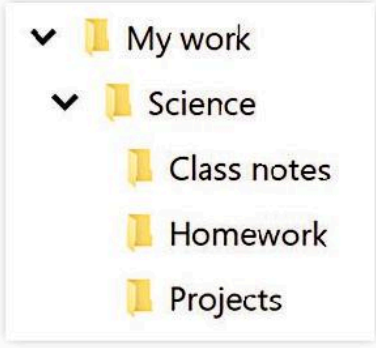
Example: You have just completed a science homework assignment on big cats. The assignment is to be handed in on 24 April 2020.
A good file name is:

- Science_homework_big_cats_24_April

Rule 2: Use folders to organise your work

Using folders to organise your files can help you to find your work. Folders also allow you to use shorter file names. You can create a folder to hold all your science work. Now your big cats file name can be shortened to:

- Homework_big_cats_24_April





Activity

There is a folder called 'Homework' shown in the directory structure. How can the file name 'Homework_big_cats_24_April' be shortened if the file is stored in that folder?

Rule 3: Save your work often

Get used to saving your work regularly and often. Don't wait until you have finished a piece of work before saving it. There will be times in between typing that you pause to think about what you will write next. That is a good time to save your work.

Rule 4: Be careful not to overwrite your files

Sometimes you will see a warning on screen like the one opposite. The warning tells you that you are about to replace a file you have previously saved. You might be about to lose important work.

If you are not sure what to do, cancel the Save and start again. If the problem continues, ask your teacher for help.



Rule 5: Make copies of important files

Sometimes you have a file that is so important that losing it would be a disaster. For example, it might be an important project that has taken you a long time to complete. You can reduce the risk of losing important work by making a copy of the file. The copy is called a back-up file.



Activity

Make an information sheet to promote good practice about saving files. Use the best practice rules in this section to help you.



Extra challenge

Work with a partner. How well does your partner use file names and folders? Suggest how your partner can make improvements to their file storage.



Explore more

Spend time at home with a family member who has work stored on a computer or tablet. How do they organise their data files? Can you suggest ways to improve how they organise their files? Can you learn anything from the way they work?

Check what you know

You have learned

- that computer power improves how devices, such as TVs and cars, work
- how computers and technology improve the way people work
- about computer storage and why it is important.

Test

1. You have just finished a project on your computer. Make a list of the steps you take to save your file safely.
2. You will find a microprocessor inside a tablet computer. What does the microprocessor do? Name two other types of computer that contain a microprocessor.
3. Name something that is built by robots.
4. Why is it important to save your computer work regularly?
5. List two devices used in the home and two devices used at work that use microprocessors.
6. List three ways that computers and microprocessors help doctors to do their jobs.
7. There are several ways you can save work at home so that you can use it later at school. List as many ways as you can think of. Which is the best method – and why?
8. Name a device used in the home that uses an embedded microprocessor. How does it make the device better to use?
9. How are microprocessors used to help a person drive a car?



Activities

This is a group activity. In your group, do the following.

1 Discuss how you have seen your teacher use computers and technology to do their job.

2 Write two or three questions to ask your teacher about how computer technology helps them do their job.

3 Interview your teacher. Make your own notes on how computers help teachers.

Self-evaluation

- I answered test questions 1–3.
- I completed activity 1. I discussed how teachers use computers and technology.
- I answered test questions 1–6.
- I completed activities 1 and 2. I wrote questions about computer technology to ask my teacher.
- I answered all the test questions.
- I completed all the activities.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?